

LPDDR4 高温参数方案

1.参考协议

DDR控制器可通过读取MR4的OP[2:0]获取当前所需的DDR刷新比例,常温下刷新比例为1,当出现较低温度时,刷新率需要配置为原先的4X/2X,当出现较高温度的时,则需要配置为原先刷新率的0.5/0.25

理论上,当DDR颗粒超过85℃时,推荐的刷新率为原先的0.25

1.1 读取推荐刷新率

Table 32: MR4 Op-Code Bit Definitions

Feature	Type	OP	Definition	Notes
Refresh rate	Read-only	OP[2:0]	000b: SDRAM low temperature operating limit exceeded 001b: 4x refresh 010b: 2x refresh 011b: 1x refresh (default) 100b: 0.5x refresh 101b: 0.25x refresh, no derating 110b: 0.25x refresh, with derating 111b: SDRAM high temperature operating limit exceeded	1-4, 7-9
SR abort (Self refresh abort)	Write	OP[3]	0b: Disable (default) 1b: Device dependent	9
PPRE (Post-package repair entry/exit)	Write	OP[4]	0b: Exit PPR mode (default) 1b: Enter PPR mode (Reference MR25 OP[7:0] for available PPR resources)	5, 9
Thermal offset-controller offset to TCSR	Write	OP[6:5]	00b: No offset, 0-5°C gradient (default) 01b: 5°C offset, 5-10°C gradient 10b: 10°C offset, 10-15°C gradient 11b: Reserved	9
TUF (Temperature update flag)	Read-only	OP7	0b: OP[2:0] No change in OP[2:0] since last MR4 read (default) 1b: Change in OP[2:0] since last MR4 read	6-8

- Notes:
1. The refresh rate for each MR4 OP[2:0] setting applies to t_{REFI} , t_{REFIpb} , and t_{REFW} . MR4 OP[2:0] = 011b corresponds to a device temperature of 85°C. Other values require either a longer (2x, 4x) refresh interval at lower temperatures or a shorter (0.5x, 0.25x) refresh interval at higher temperatures. If MR4 OP[2] = 1b, the device temperature is greater than 85°C.
 2. At higher temperatures (>85°C), AC timing derating may be required. If derating is required the device will set MR4 OP[2:0] = 110b. See derating timing requirements in the AC Timing section.
 3. DRAM vendors may or may not report all of the possible settings over the operating temperature range of the device. Each vendor guarantees that their device will work at any temperature within the range using the refresh interval requested by their device.
 4. The device may not operate properly when MR4 OP[2:0] = 000b or 111b.
 5. Post-package repair can be entered or exited by writing to MR4 OP[4].
 6. When MR4 OP[7] = 1b, the refresh rate reported in MR4 OP[2:0] has changed since the last MR4 read. A mode register read from MR4 will reset MR4 OP[7] to 0b.
 7. MR4 OP[7] = 0b at power-up. MR4 OP[2:0] bits are valid after initialization sequence (Te).
 8. See the Temperature Sensor section for information on the recommended frequency of reading MR4.

1.2 关于更新刷新参数的频率

协议推荐至少200us更新一次刷新参数

Thermal Offset

Because of tight thermal coupling, hot spots on an SOC can induce thermal gradients across the device. Because these hot spots may not be located near the thermal sensor, the temperature compensated self refresh (TCSR) circuit may not generate enough refresh cycles to guarantee memory retention. To address this shortcoming, the controller can provide a thermal offset that the memory can use to adjust its TCSR circuit to ensure reliable operation.

This thermal offset is provided through MR4 OP[6:5] to either or both channels (dual-channel devices). This temperature offset may modify refresh behaviour for the channel to which the offset is provided. It will take a maximum of 200 μ s to have the change reflected in MR4 OP[2:0] for the channel to which the offset is provided. If the induced thermal gradient from the device temperature sensor location to the hot spot location of the controller is greater than 15°C, self refresh mode will not reliably maintain memory contents.

To accurately determine the temperature gradient between the memory thermal sensor and the induced hot spot, the memory thermal sensor location must be provided to the controller.

2.DDR控制器实现

DDR控制器支持通过硬件实现定时自动访问MR4寄存器,实现对刷新定时的更新

14.5.2 Automatic MR4 Polling

Alternatively, the Cadence DDR Controller has implemented automatic polling of MR4. The FM module determines how often mode register reads to MR4 will occur to gather temperature or refresh rate information. When the MR4 mode register is read, bits [7] and [4:0] of MR4 will be loaded into the `auto_tempchk_val_X` parameter.

2.1 DDR控制器刷新参数更新验证

通过读取DDR控制器内部刷新相关的参数,观测DDR刷新参数是否被正确更新

```
reg_value = read_register(0x0000014c) & ~(0xfff<<16)| (0x200<<16);  
write_register(0x0000014c, reg_value); //tREFI_CS0:0x410  
printf("tREFI_CS0  %x\n", reg_value);  
  
reg_value = read_register(0x00000164) & ~(0xfff<<16)| (0x200<<16);  
write_register(0x00000164, reg_value); //tREFI_CS1:0x410  
printf("tREFI_CS1  %x\n", reg_value);
```